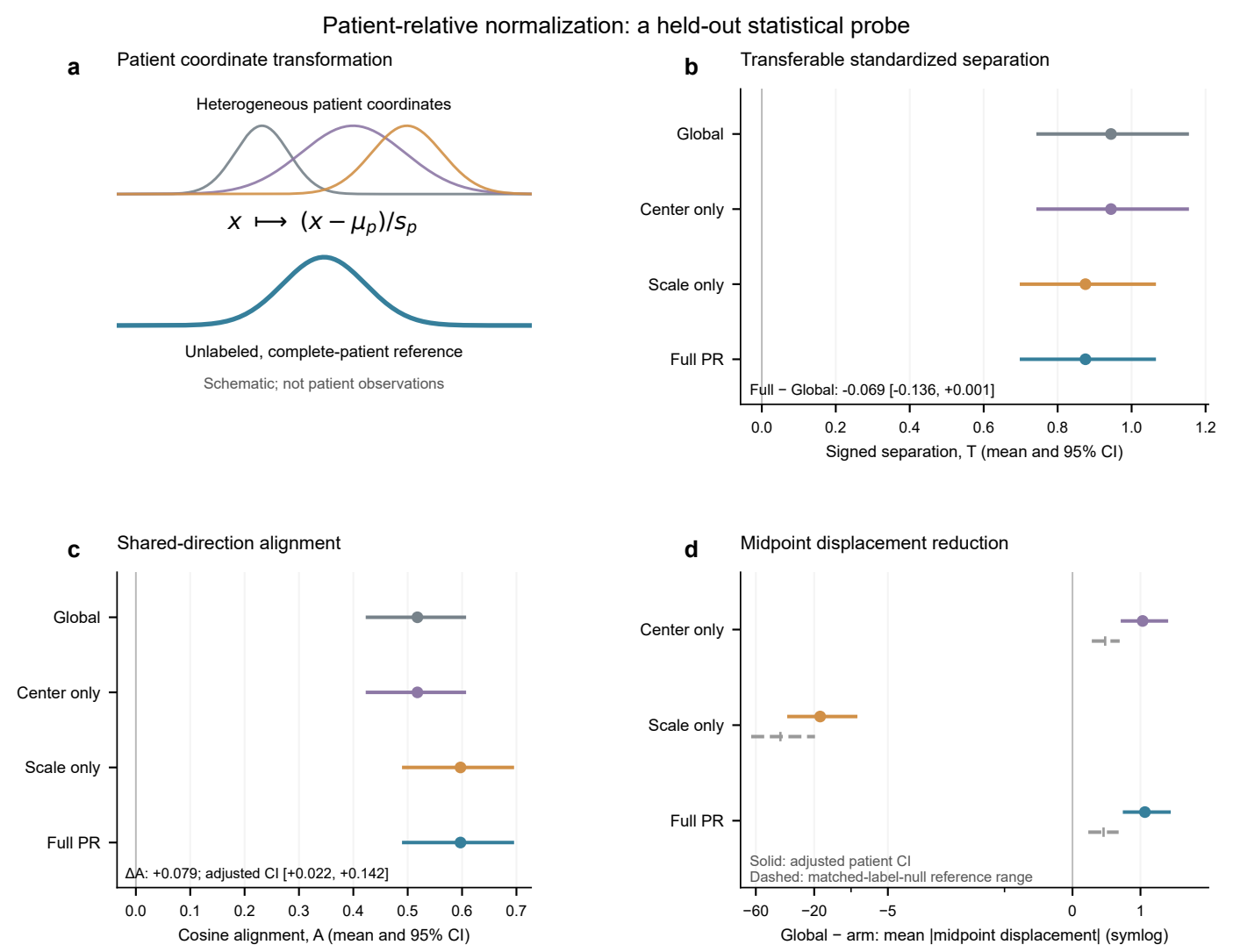

Patient-relative normalization: a held-out statistical probe
a
Patient coordinate transformation
Heterogeneous patient coordinates
$x \mapsto (x - \mu_p)/s_p$
Unlabeled, complete-patient reference
Schematic; not patient observations
b
Transferable standardized separation
Global
Center only
Scale only
Full PR
Full − Global: -0.069 [-0.136, +0.001]
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Signed separation, T (mean and 95% CI)
c
Shared-direction alignment
Global
Center only
Scale only
Full PR
ΔA: +0.079; adjusted CI [+0.022, +0.142]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Cosine alignment, A (mean and 95% CI)
d
Midpoint displacement reduction
Center only
Scale only
Full PR
Solid: adjusted patient CI
Dashed: matched-label-null reference range
−60
−20
−5
0
1
Global − arm: mean |midpoint displacement| (symlog)
n = 80 unique patients; one held-out assessment per patient; fixed TRAIN-derived directions